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Some topics in commutative algebra

Unless otherwise specified, all rings in this book will be supposed commutative and with unit.

In this chapter, we introduce some indispensable basic notions of commutative algebra such as the tensor product, localization, and flatness. Other, more elaborate notions will be dealt with later, as they are needed. We assume that the reader is familiar with linear algebra over a commutative ring, and with Noetherian rings and modules.

1.1 Tensor products

In the theory of schemes, the fibered product plays an important role (in particular the technique of base change). The corresponding notion in commutative algebra is the tensor product of modules over a ring.

1.1.1 Tensor product of modules

Definition 1.1. Let A be a commutative ring with unit. Let M, N be two A-modules. The tensor product of M and N over A is defined to be an A-module H, together with a bilinear map $\phi : M \times N \to H$ satisfying the following universal property:

For every A-module L and every bilinear map $f : M \times N \to L$, there exists a unique homomorphism of A-modules $\tilde{f} : H \to L$ making the following diagram commutative:

$$\begin{array}{ccc} M \times N & \overset{f}{\to} & L \\ \phi \searrow & & \nearrow \tilde{f} \\ & H & \end{array}$$

Proposition 1.2. Let A be a ring, and let M, N be A-modules. The tensor product (H, ϕ) exists, and is unique up to isomorphism.

Proof. As the solution of a universal problem, the uniqueness is automatic, and its proof is standard. We give it here as an example. Let (H, ϕ) and (H', ϕ') be two solutions. By the universal property, ϕ and ϕ' factor respectively as $\phi = \tilde{\phi} \circ \phi'$ and $\phi' = \tilde{\phi}' \circ \phi$. It follows that $\phi = (\tilde{\phi} \circ \tilde{\phi}') \circ \phi$. As $\phi = \mathrm{Id} \circ \phi$, it follows from the uniqueness of the decomposition of ϕ that $(\tilde{\phi} \circ \tilde{\phi}') = \mathrm{Id}$. Thus we see that $\tilde{\phi} : H \to H'$ is an isomorphism.

Let us now show existence. Consider the free A-module $A^{(M \times N)}$ with basis $M \times N$. Let $\{e_{x,y}\}_{(x,y) \in M \times N}$ denote its canonical basis. Let L be the submodule of $A^{(M \times N)}$ generated by the elements having one of the following forms:

$$\begin{cases} e_{x_1+x_2,y} - e_{x_1,y} - e_{x_2,y} \\ e_{x,y_1+y_2} - e_{x,y_1} - e_{x,y_2} \\ e_{ax,y} - e_{x,ay},\ ae_{x,y} - e_{ax,y},\ a \in A \end{cases}$$

Let $H = A^{(M \times N)}/L$, and $\phi : M \times N \to H$ be the map defined by $\phi(x, y) =$ the image of $e_{x,y}$ in H. One immediately verifies that the pair (H, ϕ) verifies the universal property mentioned above.

Notation. We denote the tensor product of M and N by $(M \otimes_A N, \phi)$. In general, the map ϕ is omitted in the notation. For any $(x, y) \in M \times N$, we let $x \otimes y$ denote its image by ϕ. By the bilinearity of ϕ, we have $a(x \otimes y) = (ax) \otimes y = x \otimes (ay)$ for every $a \in A$.

Remark 1.3. By construction, $M \otimes_A N$ is generated as an A-module by its elements of the form $x \otimes y$. Thus every element of $M \otimes_A N$ can be written (though not in a unique manner) as a finite sum $\sum_i x_i \otimes y_i$, with $x_i \in M$ and $y_i \in N$. In general, an element of $M \otimes_A N$ cannot be written $x \otimes y$.

Example 1.4. Let $A = \mathbb{Z}$, $M = A/2A$, and $N = A/3A$. Then $M \otimes_A N = 0$. In fact, for every $(x, y) \in M \times N$, we have $x \otimes y = 3(x \otimes y) - 2(x \otimes y) = x \otimes (3y) - (2x) \otimes y = 0$.

Proposition 1.5. Let A be a ring, and let M, N, M_i be A-modules. We have the following canonical isomorphisms of A-modules:

- (a) $M \otimes_A A \simeq M$;
- (b) (commutativity) $M \otimes_A N \simeq N \otimes_A M$;
- (c) (associativity) $(L \otimes_A M) \otimes_A N \simeq L \otimes_A (M \otimes_A N)$;
- (d) (distributivity) $(\bigoplus_{i \in I} M_i) \otimes_A N \simeq \bigoplus_{i \in I} (M_i \otimes_A N)$.

Proof. Everything follows from the universal property. Let us, for example, show (a) and (d).

- (a) Let $\phi : M \times A \to M$ be the bilinear map defined by $(x, a) \mapsto ax$. For any bilinear map $f : M \times A \to L$, set $\tilde{f} : M \to L$, $x \mapsto f(x, 1)$. Then $f = \tilde{f} \circ \phi$, and $\tilde{f}$ is the unique linear map $M \to L$ having this property. Hence (M, ϕ) is the tensor product of M and A.
- (d) Let $\phi : (\bigoplus_{i \in I} M_i) \times N \to \bigoplus_{i \in I} (M_i \otimes_A N)$ be the map defined by

$$\phi : \left(\sum_i x_i, y \right) \mapsto \sum_i (x_i \otimes y)$$

Let $f : (\bigoplus_{i \in I} M_i) \times N \to L$ be a bilinear map. For every $i \in I$, f induces a bilinear map $f_i : M_i \times N \to L$ which factors through $\bar{f}_i : M_i \otimes_A N \to L$. One verifies that f factors uniquely as $f = \tilde{f} \circ \psi$, where $\psi : (\bigoplus_{i \in I} M_i) \times N \to (\bigoplus_{i \in I} M_i) \otimes N$ is the canonical map and $\tilde{f} = \bigoplus_i \bar{f}_i$. Hence $\bigoplus_{i \in I} (M_i \otimes_A N)$ is the tensor product of $(\bigoplus_{i \in I} M_i)$ with N.

Corollary 1.6. Let M be a free A-module with basis $\{e_i\}_{i \in I}$. Then every ele- ment of $M \otimes_A N$ can be written uniquely as a finite sum $\sum_i e_i \otimes y_i$, with $y_i \in N$. In particular, if A is a field and $\{e_i\}_{i \in I}$ (resp. $\{d_j\}_{j \in J}$) is a basis of M (resp. of N), then $\{e_i \otimes d_j\}_{(i,j) \in I \times J}$ is a basis of $M \otimes_A N$.

Remark 1.7. The associativity of the tensor product allows us to define the tensor product $M_1 \otimes_A \cdots \otimes_A M_n$ of a finite number of A-modules. This tensor product has a universal property analogous to that of the tensor product of two modules, with the bilinear maps replaced by multilinear ones.

Definition 1.8. Let $u : M \to M'$, $v : N \to N'$ be linear maps of A-modules. By the universal property of the tensor product, there exists a unique A-linear map $u \otimes v : M \otimes_A N \to M' \otimes_A N'$ such that $(u \otimes v)(x \otimes y) = u(x) \otimes v(y)$. In fact, the map $g : M \times N \to M' \otimes_A N'$ defined by $g(x, y) = u(x) \otimes v(y)$ is clearly bilinear, and hence factors uniquely as $(u \otimes v) \circ \phi$, where ϕ is the canonical map $M \times N \to M \otimes N$. The map $u \otimes v$ is called the tensor product of u and v. The notation is justified by Exercise 1.2.

Let $\rho : A \to B$ be a ring homomorphism, and N a B-module. Then ρ induces, in a natural way, the structure of an A-module on N: for any $a \in A$ and $y \in N$, we set $a \cdot y = \rho(a)y$. We denote this A-module by $\rho_* N$, or simply by N.

Definition 1.9. Let M be an A-module. We can endow $M \otimes_A N$ with the structure of a B-module as follows. Let $b \in B$. Let $t_b : N \to N$ denote the multiplication by b, and for any $z \in M \otimes_A N$, set $b \cdot z := (\mathrm{Id}_M \otimes t_b)(z)$. One easily verifies that this defines the structure of a B-module. We denote the B- module $M \otimes_A B$ by $\rho_* M$. This is called the extension of scalars of M by B. By construction, we have $b(x \otimes y) = x \otimes (by)$ for every $b \in B$, $x \in M$, and $y \in N$.

Proposition 1.10. Let $\rho : A \to B$ be a ring homomorphism, M an A-module, and let N, P be B-modules. Then there exists a canonical isomorphism of B- modules

$$M \otimes_A (N \otimes_B P) \simeq (M \otimes_A N) \otimes_B P$$

Proof. Let us show that there exist A-linear maps $f : M \otimes_A (N \otimes_B P) \to (M \otimes_A N) \otimes_B P$, $g : (M \otimes_A N) \otimes_B P \to M \otimes_A (N \otimes_B P)$ such that for every $x \in M$, $y \in N$, and $z \in P$, we have $f(x \otimes (y \otimes z)) = (x \otimes y) \otimes z$ and

$g((x \otimes y) \otimes z) = x \otimes (y \otimes z)$. This will imply that f is an isomorphism, with inverse g. The B-linearity of f follows from this identity.

Let us fix $x \in M$. Let $t_x : N \to M \otimes_A N$ denote the A-linear map defined by $t_x(y) = x \otimes y$. Consider the map $h : M \times (N \otimes_B P) \to (M \otimes_A N) \otimes_B P$ defined by $h(x, u) = (t_x \otimes \mathrm{Id}_P)(u)$. This map is A-bilinear, and hence induces an A-linear map f as desired. The construction of g is similar.

Taking $N = B$ in the proposition above, we obtain:

Corollary 1.11. Let $\rho : A \to B$ be a ring homomorphism, let M be an A- module, and N a B-module. There exists a canonical isomorphism of B-modules

$$(M \otimes_A B) \otimes_B N \simeq M \otimes_A N \quad \text{(simplification by } B\text{)}$$

1.1.2 Right-exactness of the tensor product

Let us recall that a complex of A-modules consists of a (finite or infinite) sequence of A-modules M_i, together with linear maps $f_i : M_i \to M_{i+1}$, such that $f_{i+1} \circ f_i = 0$. A complex is written more visually as

$$\cdots \to M_i \xrightarrow{f_i} M_{i+1} \xrightarrow{f_{i+1}} M_{i+2} \to \cdots$$

The complex is called exact if $\mathrm{Ker}(f_{i+1}) = \mathrm{Im}(f_i)$ for all i. An exact complex is also called an exact sequence. For example, a sequence

$$0 \to M \xrightarrow{f} N \quad \text{(resp. } M \xrightarrow{f} N \to 0\text{)}$$

is exact if and only if f is injective (resp. surjective).

Let $f : N' \to N$ be a linear map of A-modules. For simplicity, for any A- module M, we denote the linear map $f \otimes \mathrm{Id}_M : N' \otimes_A M \to N \otimes_A M$ by f_M.

Proposition 1.12. Let A be a ring, and let

$$N' \xrightarrow{f} N \xrightarrow{g} N'' \to 0$$

be an exact sequence of A-modules. Then for any A-module M, the sequence

$$N' \otimes_A M \xrightarrow{f_M} N \otimes_A M \xrightarrow{g_M} N'' \otimes_A M \to 0$$

is exact.

Proof. The surjectivity of g_M follows from that of g (use Remark 1.3). It remains to show that $\mathrm{Ker}(g_M) = \mathrm{Im}\ f_M$; in other words, that the canonical homomorphism $\bar{g} : (N \otimes_A M)/(\mathrm{Im}\ f_M) \to N'' \otimes_A M$ is an isomorphism. Let $h : N'' \times M \to (N \otimes_A M)/(\mathrm{Im}\ f_M)$ be defined by $h(x, z) =$ the image of $y \otimes z$ in the quotient, where $y \in g^{-1}(x)$. The map h is well defined and moreover bilinear. It therefore induces a linear map $\bar{h} : N'' \otimes_A M \to (N \otimes_A M)/(\mathrm{Im}\ f_M)$, and it is easy to see that this is the inverse of $\bar{g}$.

Corollary 1.13. Let A be a ring. Let N, M be A-modules, and $i : N' \to N$ a submodule of N. There exists a canonical isomorphism

$$(N \otimes_A M)/(\mathrm{Im}\ i_M) \simeq (N/N') \otimes_A M$$

In particular, if I is an ideal of A, then we have $M \otimes_A (A/I) \simeq M/(IM)$.

Proof. It suffices to apply Proposition 1.12 to the exact sequence $N' \to N \to (N/N') \to 0$. If $N = A$ and $N' = I$, we also use Proposition 1.5(a).

1.1.3 Tensor product of algebras

An A-algebra is a commutative ring B with unit, endowed with a ring homomor- phism $A \to B$. Let B and C be A-algebras. We can canonically endow $B \otimes_A C$ with the structure of an A-algebra, as follows. Let $g : B \times C \times B \times C \to B \otimes_A C$ be defined by $g(b, c, b', c') = (bb') \otimes (cc')$. This is a multilinear map, and hence factors through $\bar{g} : B \otimes_A C \otimes_A B \otimes_A C \to B \otimes_A C$ (see Remark 1.7). We can then define the product on $B \otimes_A C$ using the composition of $(B \otimes_A C) \times (B \otimes_A C) \to (B \otimes_A C) \otimes (B \otimes_A C)$ with $\bar{g}$. More precisely, we set $(\sum_i b_i \otimes c_i) \cdot (\sum_j b'_j \otimes c'_j) = \sum_{i,j} (b_i b'_j) \otimes (c_i c'_j)$. The point was to see that this is well defined.

We have homomorphisms of A-algebras $p_1 : B \to B \otimes_A C$, $p_2 : C \to B \otimes_A C$ defined by $p_1(b) = b \otimes 1$ and $p_2(c) = 1 \otimes c$. The following proposition can be verified immediately:

Proposition 1.14. Let us keep the notation above. The triplet $(B \otimes_A C, p_1, p_2)$ satisfies the following universal property:

For every A-algebra D, and for every pair of homomorphisms of A-algebras $q_1 : B \to D$, $q_2 : C \to D$, there exists a unique homo- morphism of A-algebras $q : B \otimes_A C \to D$ such that $q_i = q \circ p_i$.

Example 1.15. Let $A[T_1, \dots, T_n]$ be the polynomial ring in n variables over A. Let B be an A-algebra. One easily verifies (either with the universal property given above or using Proposition 1.5(d)) that the homomorphism of B-algebras

$$A[T_1, \dots, T_n] \otimes_A B \to B[T_1, \dots, T_n]$$

defined by $(\sum_\nu a_\nu T^\nu) \otimes b \mapsto \sum_\nu (b a_\nu T^\nu)$ is an isomorphism. In particular, taking a polynomial ring for B, we obtain

$$A[T_1, \dots, T_n] \otimes_A A[S_1, \dots, S_m] \simeq A[T_1, \dots, T_n, S_1, \dots, S_m]$$

Exercises 1.1. Let $\{M_i\}_{i \in I}$, $\{N_j\}_{j \in J}$ be two families of modules over a ring A. Show that

$$(\bigoplus_{i \in I} M_i) \otimes_A (\bigoplus_{j \in J} N_j) \simeq \bigoplus_{(i,j) \in I \times J} (M_i \otimes_A N_j)$$

1.2. Show that there exists a unique A-linear map

$$f : \mathrm{Hom}_A(M, M') \otimes_A \mathrm{Hom}_A(N, N') \to \mathrm{Hom}_A(M \otimes_A N, M' \otimes_A N')$$

such that $f(u \otimes v) = u \otimes v$.

1.3. Let M, N be A-modules, and $i : M' \to M$, $j : N' \to N$ submodules of M and N, respectively. Then there exists a canonical isomorphism

$$(M/M') \otimes_A (N/N') \simeq (M \otimes_A N)/(\mathrm{Im}\ i_N + \mathrm{Im}\ j_M)$$

Show that $(\mathbb{Z}/n\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/m\mathbb{Z}) = \mathbb{Z}/l\mathbb{Z}$, where $l = \gcd(m, n)$.

1.4. Let M, N be A-modules, and let B, C be A-algebras. - (a) If M and N are finitely generated over A, then so is $M \otimes_A N$. - (b) If B and C are finitely generated over A, then so is $B \otimes_A C$. - (c) Taking $A = \mathbb{Z}$, $M = B = \mathbb{Z}/2\mathbb{Z}$, and $N = C = \mathbb{Q}$, show that the converse of (a) and (b) is false.

1.5. Let $\rho : A \to B$ be a ring homomorphism, M an A-module, and N a B-module. Show that there exists a canonical isomorphism of A-modules

$$\mathrm{Hom}_A(M, \rho_* N) \simeq \mathrm{Hom}_B(\rho_* M, N)$$

1.6. Let $(N_i)_{i \in I}$ be a direct system of A-modules. Then for any A-module M, there exists a canonical isomorphism $\varinjlim (N_i \otimes_A M) \simeq (\varinjlim N_i) \otimes_A M$. (Hint: show that $\varinjlim (N_i \otimes_A M)$ verifies the universal property of the tensor product $\varinjlim N_i \otimes_A M$.)

1.7. Let B be an A-algebra, and let M, N be B-modules. Show that there exists a canonical surjective homomorphism $M \otimes_A N \to M \otimes_B N$.

1.2 Flatness

The flatness of a module over a ring is a property concerning extensions of scalars. In algebraic geometry, it assures a certain 'continuity' behavior. In this section, we study a few elementary aspects of flatness. We conclude the section with faithful flatness.

1.2.1 Left-exactness: flatness

Definition 2.1. An A-module M is called flat (over A) if for every injective homomorphism of A-modules $N \to N'$, $N \otimes_A M \to N' \otimes_A M$ is injective. An A-algebra B is called flat if B is flat over A for its A-module structure, and the canonical homomorphism $A \to B$ will be called a flat homomorphism.

It follows from Proposition 1.12 that if $0 \to N' \to N \to N'' \to 0$ is an exact sequence of A-modules, and M is flat over A, then

$$0 \to N' \otimes_A M \to N \otimes_A M \to N'' \otimes_A M \to 0$$

is an exact sequence.

Proposition 2.2. Let A be a ring. We have the following properties:

- (a) Every free A-module is flat.
- (b) (Product) The tensor product of modules that are flat over A is flat over A.
- (c) (Base change) Let B be an A-algebra. If M is flat over A, then $M \otimes_A B$ is flat over B.
- (d) (Transitivity) Let B be a flat A-algebra. Then every B-module that is flat over B is flat over A.

Proof. These easily follow from the definition and the general properties of the tensor product (Proposition 1.5 and Corollary 1.11).

Example 2.3. Let $A = \mathbb{Z}$, $n \geq 2$, and $M = A/nA$. Then M is not flat over A. In fact, if we tensor the canonical injection $nA \to A$ by M, the image of $nA \otimes_A M$ in $A \otimes_A M = M$ is equal to $nM = 0$, while $nA \otimes_A M \simeq A \otimes_A M = M \neq 0$.

Theorem 2.4. Let M be an A-module. Then M is flat if and only if for every ideal I of A, the canonical homomorphism $I \otimes_A M \to IM$ is an isomorphism.

Proof. If M is flat over A, then for any ideal I of A, we can tensor the canonical injection $I \hookrightarrow A$ by M. This shows that $I \otimes_A M \to A \otimes_A M = M$ is injective. The image of this map is clearly IM, whence the isomorphism $I \otimes_A M \simeq IM$.

Conversely, let us suppose that we have this isomorphism for every ideal I, and let us show that M is flat. Let $N' \to N$ be an injective homomorphism of A-modules. We need to show that $N' \otimes M \to N \otimes M$ is injective.

Let us first suppose that N is free of finite rank, and let us show the injectivity by induction on the rank n of N. The case $n = 1$ follows from the hypothesis. Let us suppose that $n \geq 2$ and that the result holds for every free module of rank $< n$. The module N is a direct sum of two free submodules N_1 and N_2, different from N. Let $N_1' = N_1 \cap N'$, and let N_2' be the image of N' in $N_2 = N/N_1$. We then have the following commutative diagram

$$\begin{array}{ccccc} N_1' & \to & N' & \to & N_2' \\ \downarrow & & \downarrow & & \downarrow \\ N_1 & \to & N & \to & N_2 \end{array}$$

whose horizontal lines are exact, and whose vertical arrows are injective. Tensor- ing by M gives the commutative diagram

$$\begin{array}{ccccc} N_1' \otimes M & \to & N' \otimes M & \to & N_2' \otimes M \\ \beta \downarrow & & \downarrow & & \gamma \downarrow \\ N_1 \otimes M & \overset{\alpha}{\to} & N \otimes M & \to & N_2 \otimes M \end{array}$$

whose horizontal lines are still exact (Proposition 1.12). The map α is injective because N_1 is a direct factor of N (Proposition 1.5(d)), and β, γ are injective by the induction hypothesis applied to the N_i. It follows that $N' \otimes M \to N \otimes M$ is injective.

Let us now suppose N is free of arbitrary rank. Let N_0 be a direct factor of N of finite rank. By the above, the map $(N' \cap N_0) \otimes M \to N_0 \otimes M$ is injective. Hence so is $(N' \cap N_0) \otimes M \to N \otimes M$, since N_0 is a direct factor of N. Because for every $x \in N' \otimes M$ there exists an N_0 such that x is contained in the image of $(N' \cap N_0) \otimes M \to N' \otimes M$ (use Corollary 1.6), we see that $N' \otimes M \to N \otimes M$ is injective.

Let N now be an arbitrary A-module. There exist a free A-module L and a surjective homomorphism $p : L \to N$. Let us set $L' = p^{-1}(N')$. We have a commutative diagram

$$\begin{array}{ccccccc} \mathrm{Ker}(p) & \to & L' & \to & N' & \to 0 \\ & & \downarrow & & \downarrow & \\ \mathrm{Ker}(p) & \overset{p}{\to} & L & \to & N & \to 0 \end{array}$$

whose horizontal lines are exact, whence the commutative diagram

$$\begin{array}{ccccccc} \mathrm{Ker}(p) \otimes M & \to & L' \otimes M & \to & N' \otimes M & \to 0 \\ & & \downarrow & & \downarrow & \\ \mathrm{Ker}(p) \otimes M & \to & L \otimes M & \to & N \otimes M & \to 0 \end{array}$$

whose horizontal lines are also exact. As the middle vertical arrow is injective, this shows the injectivity of $N' \otimes M \to N \otimes M$.

Let A be an integral domain and M an A-module. An element $x \in M$ is called a torsion element if there exists a non-zero $a \in A$ such that $ax = 0$. We call M torsion-free (over A) if there is no non-zero torsion element in M. When A is a principal ideal domain, flatness can be expressed in a very simple way.

Corollary 2.5. Let A be a principal ideal domain. An A-module M is flat if and only if it is torsion-free over A.

Proof. Let $I = aA \neq 0$ be an ideal of A. Let t_a (resp. u_a) denote multiplication by a in A (resp. in M). Then $t_a : A \to I$ is an isomorphism. We have the following commutative diagram:

$$\begin{array}{ccc} M = A \otimes_A M & \xrightarrow{t_a \otimes \mathrm{Id}_M} & I \otimes_A M \\ & \searrow_{u_a} & \downarrow_f \\ & & IM \end{array}$$

where f is the canonical homomorphism. Consequently, f is an isomorphism if and only if u_a is an isomorphism, which is equivalent to saying that $ax = 0$ for $x \in M$ implies that $x = 0$. Hence the corollary follows from Theorem 2.4.

Proposition 2.6. Let A be a ring. Let $0 \to M' \to M \to M'' \to 0$ be an exact sequence of A-modules. Let us suppose that M'' is flat. Then the sequence $0 \to M' \otimes N \to M \otimes N \to M'' \otimes N \to 0$ is also exact for any A-module N.

Proof. It suffices to show the injectivity of $M' \otimes N \to M \otimes N$ (Proposition 1.12). Let us write N as the quotient of a free A-module L by a submodule K. We then have a commutative diagram of exact sequences:

$$\begin{array}{ccccccc} M' \otimes K & \to & M \otimes K & \to & M'' \otimes K & & \\ \downarrow & & \downarrow & & \downarrow_\alpha & & \\ 0 \to M' \otimes L & \to & M \otimes L & \to & M'' \otimes L & \to 0 & \\ & \downarrow_\beta & & \downarrow & & \downarrow & \\ & M' \otimes N & \to & M \otimes N & \to & M'' \otimes N & \end{array}$$

with α injective (M'' is flat) and β surjective. Some diagram chasing now imme- diately implies that $M' \otimes N \to M \otimes N$ is injective.

1.2.2 Local nature of flatness

Let A be a ring. The intersection of all maximal ideals of A is denoted by $\mathrm{Rad}(A)$. If A is a local ring, i.e. a ring having only one maximal ideal $\mathfrak{m}$, then $\mathrm{Rad}(A) = \mathfrak{m}$. The following lemma is very useful:

Lemma 2.7 (Nakayama's lemma). Let A be a ring, $I = \mathrm{Rad}(A)$ and M a finitely generated A-module such that $M = IM$. Then $M = 0$.

Proof. Let $\{x_1, \ldots, x_n\}$ be a system of generators of M. We may suppose n minimal. There exist $\alpha_i \in I$ such that $x_n = \sum_{i<n} \alpha_i x_i$. Hence $(1 - \alpha_n)x_n = \sum_{i<n} \alpha_i x_i$. As $1 - \alpha_n$ is invertible, and n is assumed to be minimal, it follows that $n = 1$ and $x_n = 0$.

Remark 2.8. Let us note that following Corollary 1.13, the condition $M = IM$ is equivalent to $M \otimes_A A/I = 0$.

Corollary 2.9. Let A be a ring, $I = \mathrm{Rad}(A)$, M a finitely generated A-module, and N a submodule of M. Suppose that $M \subseteq N + IM$. Then $M = N$.

Proof. In fact, M/N is a finitely generated A-module, and $(M/N) = I(M/N)$. Hence $M/N = 0$.

Some properties of the localization of modules Let A be a ring, S a multiplicative subset of A (i.e. a subset of A that is stable for multiplication, and which contains the unit element of A), and M an A- module. We define the localization $S^{-1}M$ of M with respect to S as follows. It is the set of elements written (formally) m/s, with $m \in M$ and $s \in S$, modulo the equivalence relation $m/s \sim m'/s' \iff$ there exists an $s'' \in S$ such that $s''(s'm - sm') = 0$. We endow $S^{-1}M$ with the structure of a commutative group by setting

$$m/s + m'/s' = (s'm + sm')/(ss')$$

Note that if $0 \in S$, then $S^{-1}M = 0$. If $M = A$, we define a ring structure on $S^{-1}A$ by setting

$$(a/s) \cdot (a'/s') = (aa')/(ss')$$

The canonical homomorphism $\rho : A \to S^{-1}A$ defined by $\rho(a) = a/1$ makes $S^{-1}A$ into an A-algebra. Moreover, the correspondences $I \mapsto I(S^{-1}A)$, $J \mapsto \rho^{-1}(J)$ establish a bijection between those prime ideals of A that are disjoint from S and the prime ideals of $S^{-1}A$. Let $\phi : A \to B$ be a ring homomorphism. Then $\phi(S) \subseteq B^*$ (invertible elements of B) if and only if ϕ factors through some A-algebra homomorphism $S^{-1}A \to B$.

For any A-module M, we have the structure of an $S^{-1}A$-module on $S^{-1}M$ by setting

$$(a/s) \cdot (m/s') = (am)/(ss')$$

For $f \in A$, we let M_f denote the localization of M with respect to the multi- plicative set $\{f^n \mid n \geq 1\} \cup \{1\}$. For a prime ideal $\mathfrak{p}$ of A, the localization of M with respect to the multiplicative set $A \setminus \mathfrak{p}$ is denoted $M_\mathfrak{p}$.

Lemma 2.10. Let A be a ring, S a multiplicative subset of A, and M an A-module. There exists a canonical isomorphism of $S^{-1}A$-modules $M \otimes_A S^{-1}A \simeq S^{-1}M$.

Proof. We have a canonical homomorphism of A-modules $M \otimes_A S^{-1}A \to S^{-1}M$ defined by $m \otimes (a/s) \mapsto (am)/s$. This is an isomorphism because it has an inverse given by $m/s \mapsto m \otimes (1/s)$ (this map is well defined). Moreover, this homomorphism is clearly compatible with the structure of $S^{-1}A$-modules.

Corollary 2.11. For any multiplicative subset S of A, the canonical homomor- phism $A \to S^{-1}A$ is flat.

Proof. Let N be a sub-A-module of M. It is clear, by construction, that $S^{-1}N$ is a submodule of $S^{-1}M$. Hence $N \otimes_A S^{-1}A \to M \otimes_A S^{-1}A$ is injective.

Lemma 2.12. Let M be an A-module. Then $M = 0$ if and only if $M_\mathfrak{m} = 0$ for every maximal ideal $\mathfrak{m}$ of A.

Proof. Let $x \in M$. Let us consider the ideal $I = \{a \in A \mid ax = 0\}$. If $I \neq A$, there exists a maximal ideal $\mathfrak{m}$ of A such that $I \subseteq \mathfrak{m}$. As $M_\mathfrak{m} = 0$, there exists an $s \in A \setminus \mathfrak{m}$ such that $sx = 0$. Hence $s \in I$, which contradicts the assumption that $I \subseteq \mathfrak{m}$. Consequently, $I = A$ and $1 \in I$, and hence $x = 0$.

Proposition 2.13. Let M be an A-module. The following properties are equi- valent:

- (i) M is flat over A.
- (ii) $M_\mathfrak{p}$ is flat over $A_\mathfrak{p}$ for every prime ideal $\mathfrak{p}$ of A.
- (iii) $M_\mathfrak{m}$ is flat over $A_\mathfrak{m}$ for every maximal ideal $\mathfrak{m}$ of A.

Proof. (i) $\Rightarrow$ (ii) follows from Lemma 2.10 and Proposition 2.2(c). As (ii) trivially implies (iii), it remains to show that (iii) implies (i). Let $N' \to N$ be an injective homomorphism. Let L denote the kernel of $N' \otimes_A M \to N \otimes_A M$. We have an exact sequence

$$0 \to L \to N' \otimes_A M \to N \otimes_A M \tag{2.1}$$

For any maximal ideal $\mathfrak{m}$ of A, we have an exact sequence

$$0 \to L \otimes_A A_\mathfrak{m} \to (N' \otimes_A M) \otimes_A A_\mathfrak{m} \to (N \otimes_A M) \otimes_A A_\mathfrak{m}$$

(by virtue of Corollary 2.11). As $(N' \otimes_A M) \otimes_A A_\mathfrak{m} = (N'_\mathfrak{m}) \otimes_{A_\mathfrak{m}} M_\mathfrak{m}$, and $N'_\mathfrak{m} \to N_\mathfrak{m}$ is injective, we have $L_\mathfrak{m} = L \otimes_A A_\mathfrak{m} = 0$. It follows by Lemma 2.12 that $L = 0$. Hence M is flat over A.

Let us recall that a Dedekind domain is a Noetherian integral domain A whose localizations $A_\mathfrak{p}$ at the prime ideals $\mathfrak{p}$ are principal ideal domains (we will see the equivalence with the usual definition in Subsection 4.1.1). The following corollary is an immediate consequence of the proposition above and Corollary 2.5.

Corollary 2.14. Let A be a Dedekind domain. An A-module is flat if and only if it is torsion-free over A. In particular, every injective ring homomorphism $A \to B$ with B an integral domain is flat.

Corollary 2.15. Let $\rho : A \to B$ be a ring homomorphism. The following prop- erties are equivalent:

- (i) $A \to B$ is flat.
- (ii) For every prime ideal $\mathfrak{q}$ of B, $B_\mathfrak{q}$ is flat over $A_\mathfrak{p}$, where $\mathfrak{p} = \rho^{-1}(\mathfrak{q})$.
- (iii) For every maximal ideal $\mathfrak{q}$ of B, $B_\mathfrak{q}$ is flat over $A_\mathfrak{p}$.

Proof. (i) implies (ii): it is easy to see that $B_\mathfrak{q}$ is a localization of (and hence is flat over) $B \otimes_A A_\mathfrak{p}$. Now, the latter is flat over $A_\mathfrak{p}$ since B is flat over A. Hence $B_\mathfrak{q}$ is flat over $A_\mathfrak{p}$ by the transitivity of flatness (Proposition 2.2(d)).

The proof of (iii) $\Rightarrow$ (i) is analogous to that of Proposition 2.13: one can replace M by B in exact sequence (2.1); then L is a B-module, and $L \otimes_B B_\mathfrak{q} = 0$ for every maximal ideal $\mathfrak{q}$ of B. Hence $L = 0$ and B is flat over A.

Let us conclude with the following theorem, which is a partial converse to Proposition 2.2(a). This converse is false in general, as is shown by the example $A = \mathbb{Z}_p\mathbb{Z}$ and $M = \mathrm{Frac}(A)$.

Theorem 2.16. Let M be a finitely generated flat module over a local ring A. Then M is free over A.

Proof. Let $\mathfrak{m}$ be the maximal ideal of A, and $k = A/\mathfrak{m}$ the residue field of A. Then $M/\mathfrak{m}M = M \otimes_A k$ is a vector space over k. Let $\{x_1, \ldots, x_n\}$ be a set of elements of M whose images in $M/\mathfrak{m}M$ form a free set over

k. Let us show that the set is free over A. Suppose that $\sum_i a_i x_i = 0$ with $a_i \in A$. Consider the linear map $f : A^n \to A$ defined by $f(b_1, \dots, b_n) = \sum_i a_i b_i$. We have an exact sequence

$$\mathrm{Ker}(f) \to A^n \xrightarrow{f} A$$

This gives an exact sequence when we tensor by M (here we use the flatness assumption on M):

$$\mathrm{Ker}(f) \otimes_A M \xrightarrow{f_M} M^n \xrightarrow{f_M} M$$

where f_M is defined by $f_M(y_1, \dots, y_n) = \sum_i a_i y_i$. Hence $(x_1, \dots, x_n) \in \mathrm{Ker}(f) \otimes M$. There exist $r \in \mathbb{N}$ and elements $b_j = (b_{1j}, \dots, b_{nj}) \in \mathrm{Ker}(f)$, $y_j \in M$, $1 \leq j \leq r$, such that $(x_1, \dots, x_n) = \sum_j b_j \otimes y_j$. At least one of the coordinates b_{ij} is not an element of $\mathfrak{m}$. We can, for example, suppose that $b_{11} \notin \mathfrak{m}$ (hence it is invertible). From the equality $\sum_i b_{i1} a_i = 0$, we deduce that

$$a_1 + a_2 c_2 + \cdots + a_n c_n = 0, \quad \text{where } c_i = b_{i1} b_{11}^{-1}$$

In particular, if $n = 1$, then $a_1 = 0$. If $n \geq 2$, we have

$$a_2(x_2 - c_2 x_1) + \cdots + a_n(x_n - c_n x_1) = 0$$

A reasoning using induction on n shows that $\{x_i\}_i$ is free.

Now let $\{x_1, \dots, x_n\}$ be a set of elements of M whose images in $M/\mathfrak{m}M$ form a basis over k. Let N be the submodule of M generated by the x_i. Then the canonical homomorphism $N \otimes k \to M \otimes k$ is surjective. Hence $(M/N) \otimes k = 0$. By Nakayama's lemma, $M/N = 0$. Hence the x_i form a basis of M.

1.2.3 Faithful flatness

Proposition 2.17. Let M be a flat A-module. Then the following properties are equivalent:

- (i) $M \neq \mathfrak{m}M$ for every maximal ideal $\mathfrak{m}$ of A.
- (ii) Let N be an A-module. If $M \otimes_A N = 0$, then $N = 0$.
- (iii) Let $f : N_1 \to N_2$ be a homomorphism of A-modules. If $f_M : N_1 \otimes M \to N_2 \otimes M$ is an isomorphism, then so is f.

Proof. (i) implies (ii): it suffices to show that for every finitely generated sub- module N_0 of N, we have $N_0 = 0$. As it is, for every submodule N_0 of N, we have $N_0 \otimes M \hookrightarrow N \otimes M$, so $N_0 \otimes M = 0$. We may assume that N is itself finitely gen- erated. Moreover, by virtue of Lemma 2.12, we may assume that A is local with maximal ideal $\mathfrak{m}$. By tensoring with $k := A/\mathfrak{m}$, we obtain $M/\mathfrak{m}M \otimes_k N/\mathfrak{m}N = 0$. It follows that $N/\mathfrak{m}N = 0$ (Corollary 1.6). Hence $N = 0$, by Nakayama's lemma.

(ii) implies (iii): we have $\mathrm{Coker}(f_M) = \mathrm{Coker}(f) \otimes M$, and since M is flat, $\mathrm{Ker}(f_M) = \mathrm{Ker}(f) \otimes M$. Hence if f_M is an isomorphism, then so is f.

Finally, (iii) implies (i) since if $M = \mathfrak{m}M$, then the constant map $0 \to A/\mathfrak{m}$ is an isomorphism after tensoring with M, so $A/\mathfrak{m} = 0$. This is impossible.

Definition 2.18. Let M be a flat module over a ring A. We say that M is faithfully flat over A if it verifies one of the properties of the proposition above. Let $f : A \to B$ be a ring homomorphism. We say that B is faithfully flat over A if it is faithfully flat as an A-module. We will also say that f is faithfully flat.

Remark 2.19. One can immediately verify that Proposition 2.2 remains true when we replace 'flat' by 'faithfully flat' and take only non-zero modules.

Corollary 2.20. Let $f : A \to B$ be a flat ring homomorphism. The following properties are equivalent:

- (i) f is faithfully flat.
- (ii) For every prime ideal $\mathfrak{p}$ of A, there exists a prime ideal $\mathfrak{q}$ of B such that $f^{-1}(\mathfrak{q}) = \mathfrak{p}$.
- (iii) For every maximal ideal $\mathfrak{m}$ of A, there exists a maximal ideal $\mathfrak{n}$ of B such that $f^{-1}(\mathfrak{n}) = \mathfrak{m}$.

Proof. Let us show the only non-trivial implication, (i) $\Rightarrow$ (ii). Let us first note that f is necessarily injective, because $\mathrm{Ker}(f) \otimes_A B \simeq \mathrm{Ker}(f)B = 0$. Since $A/\mathfrak{p} \to B/\mathfrak{p}B$ is faithfully flat (see Remark 2.19), we may assume that $\mathfrak{p} = 0$, and that A is an integral domain. Let $\rho : B \to B \otimes_A \mathrm{Frac}(A)$ be the canonical map. Let $\mathfrak{m}$ be a maximal ideal of $B \otimes_A \mathrm{Frac}(A)$. Then $\mathfrak{q} := \rho^{-1}(\mathfrak{m})$ is a prime ideal of B. Since $\rho \circ f : A \to B \otimes_A \mathrm{Frac}(A)$ factorizes into $A \to \mathrm{Frac}(A) \to B \otimes_A \mathrm{Frac}(A)$, and the inverse image of $\mathfrak{m}$ in $\mathrm{Frac}(A)$ is zero, we have $f^{-1}(\mathfrak{q}) = (\rho \circ f)^{-1}(\mathfrak{m}) = 0$.

Exercises 2.1. Let M be an A-module. We call the ideal $\{a \in A \mid aM = 0\}$ of A the annihilator of M, and we denote it by $\mathrm{Ann}(M)$. Let $I \subseteq \mathrm{Ann}(M)$ be an ideal. - (a) Show that M is endowed, in a natural way, with the structure of an A/I-module, and that $M \simeq M \otimes_A A/I$. - (b) Let N be another A-module such that $I \subseteq \mathrm{Ann}(N)$. Show that the canonical homomorphism $M \otimes_A N \to M \otimes_{A/I} N$ is an isomorphism.

2.2. Let $\rho : A \to B$ be a ring homomorphism, S a multiplicative subset of A, and $T = \rho(S)$. Show that T is a multiplicative subset of B, and that $T^{-1}B \simeq B \otimes_A S^{-1}A$ as A-algebras.

2.3. Show that Nakayama's lemma is false for modules M that are not finitely generated.

2.4. Let I be a finitely generated ideal of A. Show that the following properties are equivalent: - (i) A/I is flat over A; - (ii) $I = I^2$; - (iii) there exists an $e \in A$ such that $e^2 = e$ and $I = eA$.

2.5. Let A be an integral domain. Show that every A-module is flat if and only if A is a field.

2.6. Let B be a flat A-algebra. - (a) Show that for any finite family $\{I_\lambda\}_{\lambda\in\Lambda}$ of ideals of A, we have $\bigcap_{\lambda\in\Lambda}(I_\lambda B) = (\bigcap_{\lambda\in\Lambda} I_\lambda)B$. - (b) Let us suppose that B is faithfully flat. Show that $IB \cap A = I$ for any ideal I of A. - (c) Let k be a field, $A = k[t, s]$, and $C = A[z]/(tz - s)$. By considering the ideals tA and sA, show that C is not flat over A.

2.7. Give an example of a finitely generated flat module that is not free (over a suitable ring A).

2.8. Let A be a Noetherian ring, M a finitely generated A-module, and N an A-module. Let B be a flat A-algebra. Let us consider the canonical homomorphism

$$\rho : \mathrm{Hom}_A(M, N) \otimes_A B \to \mathrm{Hom}_B(M \otimes_A B, N \otimes_A B)$$

- (a) Show that ρ is an isomorphism if M is free of finite rank. - (b) Let $0 \to K \to L \to M \to 0$ be an exact sequence of A-modules. Show that we have a canonical exact sequence

$$0 \to \mathrm{Hom}_A(M, N) \to \mathrm{Hom}_A(L, N) \to \mathrm{Hom}_A(K, N)$$

- (c) By taking for L a free module of finite rank, show that ρ is injective. By applying the injectivity to K, show that ρ is an isomorphism.

2.9. Let A be an integral domain, and K its field of fractions. Let M be a finitely generated sub-A-module of K. Show that M is flat if and only if it is locally free of rank 1 (i.e., $M_\mathfrak{p}$ is free of rank 1 over $A_\mathfrak{p}$ for every prime ideal $\mathfrak{p}$ of A).

2.10. Let A be an integral domain, and B its integral closure in the field of fractions $\mathrm{Frac}(A)$. Suppose that B is a finitely generated A-module. Show that B is flat over A if and only if $B = A$. One can show that this result is true without the assumption of finiteness of B over A.

2.11. Let M be a flat A-module, and let $\rho : A \to B$ be a ring homomorphism. Show that the following properties are true. - (a) If A is an integral domain, then M is torsion-free. - (b) The canonical homomorphism $M \to M \otimes_A B$ is injective if and only if $\mathrm{Ker}(\rho)$ is contained in $\mathrm{Ann}(M)$.

2.12. Let B be a flat algebra over a Dedekind domain A. Let $f \in B$ be such that for every maximal ideal $\mathfrak{m}$ of A, the image of f in $B/\mathfrak{m}B$ is not a zero divisor. Show that B/fB is still flat over A. See also Lemma 4.3.16.

2.13. Let M be a faithfully flat A-module. Let $N' \to N \to N''$ be a sequence of A-modules. Show that it is exact if and only if the sequence $N' \otimes M \to N \otimes M \to N'' \otimes M$ is exact.

2.14. Let $A \to B$ be a ring homomorphism, and let J be an ideal of B such that B/J is flat over A. Show that for any ideal I of A, we have $(IB) \cap J = IJ$ (tensor the injection $I \to A$ by B/J).

2.15. Let $0 \to M' \to M \to M'' \to 0$ be an exact sequence of modules over a ring A. Let us suppose that M'' is flat. Show that M' is flat if and only if M is flat.

2.16. Let $(A, \mathfrak{m})$ be a Noetherian local ring, and

$$C^\bullet : 0 \to M' \to M \to M'' \to 0$$

a complex of finitely generated flat A-modules. Show that if there exists an ideal $I \subseteq \mathfrak{m}$ such that $C^\bullet \otimes_A A/I$ is exact, then $C^\bullet$ is exact.

2.17. Let M be a finitely generated flat A-module. Show that it is faithfully flat if and only if $\mathrm{Ann}(M) = 0$.

2.18. Let B be an A-algebra, and let E be a faithfully flat B-module. Show that E is flat over A if and only if B is flat over A.

2.19. Let $f : A \to B$ be faithfully flat ring homomorphism. - (a) Show that f is injective and that $I \to I \otimes_A B$ is injective for every ideal I of A. - (b) Let $N = \mathrm{Coker}(f)$ be the cokernel of f. Let I be an ideal of A. Using the commutative diagram

$$\begin{array}{ccccccc} I & \to & I \otimes_A B & \to & I \otimes_A N & \to 0 \\ & & \downarrow & & \downarrow & \\ & & IB & \to & IN & \end{array}$$

and Exercise 2.6(b), show that $I \otimes_A N \to IN$ is injective, and hence N is a flat A-module. - (c) Show that for any A-module M, the canonical map $M \to M \otimes_A B$ is injective.

1.3 Formal completion

1.3.1 Inverse limits and completions

Let us first recall some notions and properties of topological groups. An (Abelian) topological group is an Abelian group G endowed with the structure of a topological space for which the homomorphism $G \times G \to G$ defined by $(x, y) \mapsto x - y$ is continuous. Such a structure is entirely determined by giving a fundamental system V of neighborhoods of 0 such that every element $V \in V$ contains the difference $V_1 - V_2$ of two elements of V.

A (descending) filtration $(G_n)_n$ (i.e., a descending chain of subgroups $(G_n)_n$ of G) defines a unique structure of topological group on G for which the G_n form a fundamental system of neighborhoods of 0. In this section we are essen- tially interested in topologies of this type. For this topology, G is separated (i.e., Hausdorff) if and only if $\bigcap_n G_n = \{0\}$. Two filtrations $(G_n)_n$, $(G'_n)_n$ define the same topology on G if and only if for every n, there exists an m such that $G'_m \subseteq G_n$, and vice versa.

Let us note that a group homomorphism $f : G \to H$ between two topological groups is continuous if and only if for every neighborhood V of 0 in H, $f^{-1}(V)$ is a neighborhood of 0 in G.

A (commutative) topological ring is a ring A endowed with a topology for which the maps $A \times A \to A$ defined by $(x, y) \mapsto x - y$ and $(x, y) \mapsto xy$ are continuous. A topological A-module M is an A-module whose additive structure is a topological group, and for which the map $(a, x) \mapsto ax$ from $A \times M$ to M is continuous. Let A be a commutative ring, and I an ideal. The ideals I^n of A define a topology on A called the I-adic topology. Let M be an A-module; then the filtration $(I^n M)_n$ defines a structure of topological A-module, called the I-adic topology on M.

Let G be a topological group defined by a filtration $(G_n)_n$. A sequence $(x_m)_m$ of elements of G is called a Cauchy sequence if for every n there exists an m_0 such that $x_m - x_{m_0} \in G_n$ for every $m \geq m_0$. The topological group G is complete if every Cauchy sequence has a limit in G. One way to construct complete groups is to construct inverse limits.

An inverse system (of sets) consists of a collection of sets $(A_n)_{n \geq 0}$ and maps $\pi_n : A_{n+1} \to A_n$ for every n. The inverse limit of the $(A_n)_n$ is the set

$$\varprojlim A_n := \{(a_n)_n \in \prod_n A_n \mid a_n = \pi_n(a_{n+1}) \text{ for all } n\}$$

For every m, the projection onto the mth coordinate defines a canonical map

$$p_m : \varprojlim A_n \to A_m$$

Let B be a set, and let $f_n : B \to A_n$ be maps such that $f_n = \pi_n \circ f_{n+1}$ for every n. Then there exists a unique map $f : B \to \varprojlim A_n$ such that $f_m = p_m \circ f$. This universal property characterizes the inverse limit.

We define inverse limits of groups, of rings, and of modules in the same way, imposing that the transition maps π_n be compatible with the group, ring, or module structure. The inverse limits are then groups, rings, or modules.

A homomorphism of inverse systems $(A_n)_n \to (B_n)_n$ of Abelian groups con- sists of homomorphisms $A_n \to B_n$ that are compatible with the projection maps $A_{n+1} \to A_n$ and $B_{n+1} \to B_n$. This canonically induces a group homomorphism $\varprojlim A_n \to \varprojlim B_n$. A sequence of homomorphisms

$$0 \to (A_n)_n \to (B_n)_n \to (C_n)_n \to 0$$

of inverse systems of Abelian groups is called an exact sequence if $0 \to A_n \to B_n \to C_n \to 0$ is exact for all n.

Lemma 3.1. Let $0 \to (A_n)_n \to (B_n)_n \to (C_n)_n \to 0$ be an exact sequence of inverse systems of Abelian groups. Then $0 \to \varprojlim A_n \to \varprojlim B_n \to \varprojlim C_n$ is exact. Moreover, the last homomorphism is surjective if there exists an n_0 such that $A_{n+1} \to A_n$ is surjective for every $n \geq n_0$ (see also Exercise 3.15).

Proof. Let us check the surjectivity of $\varprojlim B_n \to \varprojlim C_n$; the rest follows imme- diately from the definition. As the inverse limit of a system does not depend on its first terms, we may assume that $n_0 = 0$. Let $\pi_n : B_{n+1} \to B_n$ denote the tran- sition homomorphism. Let $(c_n)_n \in \varprojlim C_n$. Let $b_n \in B_n$ be an arbitrary preimage of c_n. Identifying A_n with a submodule of B_n, we have $a_n := \pi_n(b_{n+1}) - b_n \in A_n$. Hence there exists an $a_{n+1} \in A_{n+1}$ such that $\pi_n(a_{n+1}) = a_n$. It follows that $\pi_n(b_{n+1} - a_{n+1}) = b_n$. Thus we can modify the b_n one after the other to obtain $\pi_n(b_{n+1}) = b_n$ for all n. Consequently, $(b_n)_n \in \varprojlim B_n$, and its image in $\varprojlim C_n$ is $(c_n)_n$.

Let G be a topological group defined by a filtration $(G_n)_n$. We then have a natural inverse system $(G/G_n)_n$. Consider $\hat{G} := \varprojlim (G/G_n)$. Let

$$\hat{G}_n := \{(a_m)_m \in \hat{G} \mid a_m = 0 \text{ for every } m \leq n\}$$

This defines a filtration $(\hat{G}_n)_n$ on $\hat{G}$ and makes the latter into a topological group. We have a natural homomorphism $\pi : G \to \hat{G}$ defined by $\pi(g) = (g_n)_n$, where g_n is the canonical image of g in G/G_n. As $\pi^{-1}(\hat{G}_n) = G_n$, π is continuous.

Let G be an Abelian topological group. The completion of G is a complete separated Abelian topological group K together with a continuous homomor- phism $\phi : G \to K$ such that every continuous homomorphism from G to a complete separated Abelian topological group factors uniquely through ϕ. We will in general omit the term 'separated' and use 'complete' to mean 'separated and complete'.

Proposition 3.2. Let us keep the notation above. Then $\pi : G \to \hat{G}$ is the completion of G. Moreover, π induces an isomorphism $G/G_n \simeq \hat{G}/\hat{G}_n$ for every n.

Proof. It is clear that $\bigcap_n \hat{G}_n = 0$. Hence $\hat{G}$ is separated. Let $(g_m)_m$ be a Cauchy sequence in $\hat{G}$. For every $r \geq 0$, there exists an n_r such that for every $n \geq n_r$, we have $g_n - g_{n_r} \in \hat{G}_r$. If we write $g_n = (g_{1n}, \dots, g_{kn}, \dots)$, this implies that $g_{in} = g_{in_r}$ for every $i \leq r$ and every $n \geq n_r$. Let $\hat{g} = (g_{1n_1}, g_{2n_2}, \dots)$. One immediately verifies that $\hat{g} \in \hat{G}$, and that this is the limit of the sequence $(g_m)_m$. Hence $\hat{G}$ is indeed complete.

Let $(g_n)_n \in \hat{G}$. If $g \in G$ is a preimage of g_m in G, then $\pi(g) - (g_n)_n \in \hat{G}_m$. Therefore $\pi(G)$ is dense in $\hat{G}$. The isomorphism $G/G_n \simeq \hat{G}/\hat{G}_n$ immediately follows from the construction.

Let us now show the factoring property. Let $f : G \to H$ be a contin- uous homomorphism to a complete Abelian topological group. Let $\alpha \in \hat{G}$. Then there exists a sequence $(\alpha_m)_m$ of elements of G such that $\pi(\alpha_m)$ tends to α. As $\pi^{-1}(\hat{G}_n) = G_n$, this implies that $(\alpha_m)_m$, and therefore $(f(\alpha_m))_m$, is a Cauchy sequence. Let β be the limit of $(f(\alpha_m))_m$. Since H is separated, $\operatorname{Ker} \pi = \bigcap_{n \geq 0} G_n \subseteq \operatorname{Ker} f$, which implies that β does not depend on the choice of the sequence $(\alpha_m)_m$. We can therefore factor f as $\tilde{f} \circ \pi$ by setting $\tilde{f}(\alpha) = \beta$. The uniqueness of $\tilde{f}$ comes from the density of $\pi(G)$, and its continuity is easy to verify.

Lemma 3.3. Let M, N be Abelian groups endowed respectively with filtrations $(M_n)_n$, $(N_n)_n$. Let $\phi : M \to N$ be a homomorphism such that for some $n_0 \geq 0$, we have $N = \phi(M) + N_{n_0}$ and $N_n = \phi(M_n) + N_{n+1}$ for every $n \geq n_0$. Then the canonical homomorphism $\hat{M} \to \hat{N}$ is surjective.

Proof. As explained in the proof of Lemma 3.1, we may assume that $n_0 = 0$. The hypotheses imply that the canonical homomorphisms $\phi_n : M/M_n \to N/N_n$ and $\operatorname{Ker} \phi_n = \phi^{-1}(N_n)/M_n \to \operatorname{Ker} \phi_{n-1} = \phi^{-1}(N_{n-1})/M_{n-1}$ are surjective. The surjectivity of $\hat{M} \to \hat{N}$ then results from Lemma 3.1 applied to the exact sequence of inverse systems $0 \to (\operatorname{Ker} \phi_n)_n \to (M/M_n)_n \to (N/N_n)_n \to 0$.

Let A be a ring endowed with the I-adic topology. We denote the completion of A by $\hat{A}$. We also call $\hat{A}$ the formal completion of A for the I-adic topology. As A is a ring, so is $\hat{A}$. Let M be an A-module. Then any submodule filtration $(M_n)_n$ of M defines the structure of a topological A-module on M. We let $\hat{M}$ denote the completion of M for the I-adic topology (i.e., that defined by the filtration $(I^n M)$). This is an $\hat{A}$-module.

Example 3.4. Let p be a prime number. The completion of $\mathbb{Z}$ for the p-adic topology is denoted $\mathbb{Z}_p$, and is called the ring of p-adic integers.

Definition 3.5. Let A be a commutative ring with unit. The ring of formal power series in one variable $A[[T]]$ is defined in the following way. Let $A^{\mathbb{N}}$ be the group of sequences with coefficients in A. To simplify, we denote a sequence $(a_n)_{n \geq 0}$ by

$$a_0 + a_1 T + a_2 T^2 + \dots$$

We endow $A^{\mathbb{N}}$ with a multiplicative law by setting

$$\left(\sum_{i \geq 0} a_i T^i\right)\left(\sum_{j \geq 0} b_j T^j\right) = \sum_{k \geq 0} c_k T^k$$

where $c_k = \sum_{i+j=k} a_i b_j$. This ring clearly contains the polynomial ring $A[T]$. We define inductively the ring of formal power series

$$A[[T_1, \dots, T_r]] = A[[T_1, \dots, T_{r-1}]][[T_r]]$$

Example 3.6. Let $B = A[T_1, \dots, T_r]$ be a polynomial ring with coefficients in a commutative ring A. Let $\mathfrak{m}$ be the ideal of B generated by $T_1, T_2, \dots, T_r$. Then the $\mathfrak{m}$-adic completion $\hat{B}$ is isomorphic to $A[[T_1, \dots, T_r]]$, the latter being endowed with the $\mathfrak{n}$-adic topology for the ideal $\mathfrak{n}$ generated by the T_i.

In fact, the following facts are very easy to verify: $A[[T_1, \dots, T_r]]$ is complete, and the canonical map $B \to A[[T_1, \dots, T_r]]$ is continuous and induces an isomor- phism $B/\mathfrak{m}^n \simeq A[[T_1, \dots, T_r]]/\mathfrak{n}^n$ for every $n \geq 1$. This implies our assertion.

Proposition 3.7. Let A be a Noetherian ring; then the ring of formal power series $A[[T_1, \dots, T_r]]$ is also Noetherian.

Proof. By induction on r, it suffices to show that $A[[T]]$ is Noetherian. Let I be an ideal of $A[[T]]$. We have to show that I is finitely generated. For each $i \geq 0$, let $J_i = \{\alpha \in A \mid \alpha T^i \in I + T^{i+1}A[[T]]\}$. This defines an ascending sequence of ideals of A. Hence there exists a $d \geq 0$ such that $J_n = J_d$ for every $n \geq d$. Since $I/(I \cap T^d A[[T]])$ is a submodule of $A[[T]]/(T^d) = A[T]/(T^d)$ which is finitely generated over A, it is enough to show that $I \cap T^d A[[T]]$ is finitely generated over $A[[T]]$. Let $f_1, \dots, f_m$ be generators of J_d, and let $F_j \in I$ be such that $F_j - f_j T^d \in T^{d+1}A[[T]]$. Then $F_j \in I \cap T^d A[[T]]$. We are going to show that $F_1, \dots, F_m$ generate $I \cap T^d A[[T]]$. We can suppose that $I \subseteq T^d A[[T]]$.

Let $F = \sum_{i \geq 0} a_i T^i \in I$. Let $q = \min\{i \geq 0 \mid a_i \neq 0\} \geq d$. Then $a_q \in J_q = J_d$. Hence there exist $b_1, \dots, b_m \in A$ such that $a_q = \sum_{1 \leq j \leq m} b_j f_j$. It follows that

$$F - \sum_{1 \leq j \leq m} (b_j T^{q-d}) F_j \in I \cap T^{q+1} A[[T]]$$

By induction on $n \geq q$, we see that F can be written as

$$F = G_q + G_{q+1} + \cdots + G_n + H_n$$

with $G_i \in T^{i-d}(F_1, \dots, F_m)$ and $H_n \in I \cap T^{n+1}A[[T]]$. It is clear that the series $G_q + G_{q+1} + \dots$ tends to an element of $(F_1, \dots, F_m)$ and that H_n tends to 0. Hence $F \in (F_1, \dots, F_m)$.

Corollary 3.8. Let A be a Noetherian ring, and let I be an ideal of A. Then the formal completion of A for the I-adic topology is a Noetherian ring.

Proof. Let $t_1, \dots, t_r$ be a system of generators of I. Let us consider the surjective homomorphism of A-algebras $\phi : B = A[T_1, \dots, T_r] \to A$ defined by $\phi(T_i) = t_i$, and endow B with the $\mathfrak{m}$-adic topology, where $\mathfrak{m}$ is the ideal generated by the T_i. For any $n \geq 1$, we have $\phi(\mathfrak{m}^n) = I^n$. It follows from Lemma 3.3 that $A[[T_1, \dots, T_r]] = \hat{B} \to \hat{A}$ is surjective. Hence $\hat{A}$ is Noetherian by the proposition above.

Let M, N be two I-adic A-modules. It is clear that the product topology on $M \oplus N = M \times N$ is also the I-adic topology. Consequently, $(M \oplus N)\hat{} = \hat{M} \oplus \hat{N}$. In particular, every isomorphism of A-modules $A^r \to L$ canonically induces an isomorphism of $\hat{A}$-modules $\hat{A}^r \to \hat{L}$. Let M be an I-adic A-module. There exists a canonical homomorphism $M \otimes_A \hat{A} \to \hat{M}$. It is in general neither injective nor surjective. However, we do have the following lemma:

Lemma 3.9. Let M be a finitely generated A-module. Then $M \otimes_A \hat{A} \to \hat{M}$ is surjective.

Proof. Let $p : L \to M$ be a surjective homomorphism of A-modules, with L free of finite rank. We have $p(I^n L) = I^n M$. It follows from Lemma 3.3 that $\hat{L} \to \hat{M}$ is surjective. It results from the commutative diagram

$$\begin{array}{ccc} L \otimes_A \hat{A} & \to & M \otimes_A \hat{A} \\ \downarrow & & \downarrow \\ \hat{L} & \to & \hat{M} \end{array}$$

that $M \otimes_A \hat{A} \to \hat{M}$ is surjective.

1.3.2 The Artin–Rees lemma and applications

Let us first make a digression towards graded rings and modules. The topological notions do not intervene at first.

Let B be a commutative ring (with unit). A grading on B consists of a decomposition of B in subgroups

$$B = \bigoplus_{d\geq 0} B_d$$

such that $B_d B_e \subseteq B_{d+e}$. The elements of B_d are called the homogeneous elements of degree d. If B is an algebra over a ring A, we also ask that the image of A in B be contained in B_0. We then call B a graded algebra over A. A polynomial ring $B = A[T_0, \dots, T_r]$, for example, is naturally graded by taking for B_d the sub-A-module generated by the monomials of degree d.

A graded B-module is a B-module E which has a decomposition in subgroups $E = \bigoplus_{d\geq 0} E_d$ such that $B_d E_e \subseteq E_{d+e}$.

Let A be a commutative ring, and I an ideal of A. Let $\mathrm{gr}_I A := \bigoplus_{d\geq 0} I^d$ denote the graded ring. We can also see $\mathrm{gr}_I A$ as the subring $\bigoplus_{d\geq 0} I^d T^d \subseteq A[T]$. Let us note that if I is finitely generated, then $\mathrm{gr}_I A$ is a finitely generated A-algebra; that is, it is the quotient of a polynomial ring $A[T_1, \dots, T_r]$ by an ideal. In fact, let $t_1, \dots, t_r$ be a system of generators of I. Then the map $A[T_1, \dots, T_r] \to \mathrm{gr}_I A$ defined by $T_i \mapsto t_i \in I^1$ is a surjective ring homomorphism. In particular, if A is Noetherian, then $\mathrm{gr}_I A$ is Noetherian because $A[T]$ is Noetherian.

Let M be an A-module. An I-filtration of M is a filtration $(M_n)_n$ of M by submodules M_n such that $IM_n \subseteq M_{n+1}$. We call the filtration stable if there exists an n_0 such that $M_{n+1} = IM_n$ for every $n \geq n_0$.

Let $\widetilde{M} := \bigoplus_{n\geq 0} M_n$. This is a graded $\mathrm{gr}_I A$-module.

Lemma 3.10. Let A be a Noetherian ring, M a finitely generated A-module, and $(M_n)_n$ an I-filtration on M. The following properties are equivalent:

- (i) The filtration $(M_n)_n$ is stable.
- (ii) The module $\widetilde{M}$ is finitely generated over $\mathrm{gr}_I A$.

Proof. Let $N_n = \bigoplus_{i\leq n} M_i$. This is a finitely generated A-module. Let

$$P_n = N_n \oplus \left(\bigoplus_{j\geq 1} I^j M_n \right) \subseteq \widetilde{M}$$

(where $I^j M_n$ is in the component M_{n+j}). Then the P_n are finitely generated A-modules, and form an ascending sequence whose union equals all of $\widetilde{M}$. As A is a Noetherian ring, $\widetilde{M}$ is finitely generated if and only if there exists an n_0 such that $P_{n+1} = P_n$ for every $n \geq n_0$. Now, this equality is equivalent to $IM_n = M_{n+1}$.

Proposition 3.11. Let A be a Noetherian ring, I an ideal of A, and M a finitely generated A-module endowed with a stable I-filtration $(M_n)_n$. Then for any submodule N of M, the I-filtration $(M_n \cap N)_n$ of N is also stable.

Proof. In fact, the filtration $(M_n \cap N)_n$ induces on N the structure of a sub- module of $\widetilde{M}$. The proposition results from Lemma 3.10.

Taking $M_n = I^n M$, we obtain the following corollary:

Corollary 3.12 (The Artin–Rees lemma). Let A be a Noetherian ring, I an ideal of A, M a finitely generated A-module, and N a submodule of M. Then there exists an n_0 such that

$$(I^{n+1}M) \cap N = I((I^n M) \cap N)$$

for every $n \geq n_0$.

An immediate application of this result is a theorem of Krull:

Corollary 3.13. Let A be a Noetherian ring, I an ideal of A, and M a finitely generated A-module. Then $\bigcap_{n\geq 0}(I^n M)$ is the set of elements $x \in M$ for which there exists an $\alpha \in I$ such that $(1+\alpha)x = 0$.

Proof. If $(1+\alpha)x = 0$ for an $\alpha \in I$, then $x = -x\alpha \in IM$. We see by induction that $x \in I^n M$ for every n. Conversely, let us suppose that $x \in \bigcap_{n\geq 0}(I^n M)$. Let us consider the submodule $N := xA$ of M. Then by the Artin–Rees lemma, there exists an integer $n \geq 1$ such that $I^n M \cap N \subseteq IN$, and hence $x \in IN = xI$, whence the result.

Let us return to topological modules.

Corollary 3.14. Let A be a Noetherian ring, I an ideal of A, and M a finitely generated A-module. Let $\hat{M}$ be the completion of M for the I-adic topology. Then the canonical homomorphism $M \otimes_A \hat{A} \to \hat{M}$ is an isomorphism. In particular, if A is complete for the I-adic topology, then so is M.

Proof. Let $0 \to N \to L \to M \to 0$ be an exact sequence of A-modules with L free of finite rank. As the filtrations $(I^n N)_n$, $((I^n L) \cap N)_n$ define the same topology on N by the Artin–Rees lemma, $0 \to \hat{N} \to \hat{L} \to \hat{M} \to 0$ is an exact sequence (Lemma 3.1). Let us consider the commutative diagram

$$\begin{array}{ccccccc} N \otimes_A \hat{A} & \to & L \otimes_A \hat{A} & \to & M \otimes_A \hat{A} & \to 0 \\ \downarrow & & \downarrow & & \downarrow \\ \hat{N} & \to & \hat{L} & \to & \hat{M} \to 0 \end{array}$$

whose lines are exact. As the vertical arrows are surjective (Lemma 3.9), the map $M \otimes_A \hat{A} \to \hat{M}$ is bijective.

Theorem 3.15. Let A be a Noetherian ring, and I an ideal of A. Then the completion $\hat{A}$ of A for the I-adic topology is a flat ring over A.

Proof. Let J be an ideal of A. By Corollary 3.14, we have $J \otimes_A \hat{A} \simeq \hat{J}$. Let us consider the canonical homomorphism $\hat{J} \to \hat{A}$. Let us show that it is injective, which will imply the flatness of $\hat{A}$ over A (Theorem 2.4).

Let $\alpha = (\alpha_n)_n$ be an element of the kernel of $\hat{J} \to \hat{A}$. Let us fix an $n \geq 1$. By virtue of the Artin–Rees lemma, there exists an $m \geq n$ such that $I^m \cap J \subseteq I^n J$. Let $\beta_m \in J$ be an element whose image in $J/I^m J$ equals α_m; then $\beta_m \in I^n J$. As α_n is the image of β_m in $J/I^n J$, it follows that $\alpha_n = 0$ and therefore $\alpha = 0$.

1.3.3 The case of Noetherian local rings

Theorem 3.16. Let $(A, \mathfrak{m})$ be a Noetherian local ring, and $\hat{A}$ its $\mathfrak{m}$-adic com- pletion. We have the following properties:

- (a) For every $n \geq 1$, we have a canonical isomorphism $A/\mathfrak{m}^n \simeq \hat{A}/\mathfrak{m}^n \hat{A}$.
- (b) $\hat{A}$ is a local ring with maximal ideal $\mathfrak{m}\hat{A}$, which is faithfully flat over A.
- (c) Let $(B, \mathfrak{n})$ be a local ring such that $A \subseteq B \subseteq \hat{A}$ and $\mathfrak{m}B = \mathfrak{n}$. Then the $\mathfrak{n}$-adic completion $\hat{B}$ is isomorphic to $\hat{A}$.

Proof. (a) The topology on $A/\mathfrak{m}^n$ induced by the $\mathfrak{m}$-adic topology on A is discrete, giving the exact sequence $0 \to \mathfrak{m}^n \to \hat{A} \to A/\mathfrak{m}^n \to 0$, and moreover $\mathfrak{m}^n \simeq \mathfrak{m}^n \hat{A}$ (Corollary 3.14 and Theorem 3.15). This proves (a).

(b) Since $\hat{A}/\mathfrak{m}\hat{A} \simeq A/\mathfrak{m}$ is a field, $\mathfrak{m}\hat{A}$ is a maximal ideal. Let $\alpha \in \hat{A}$. We can write $\alpha = a + \varepsilon$ with $a \in A$ and $\varepsilon \in \mathfrak{m}\hat{A}$. If $\alpha \notin \mathfrak{m}\hat{A}$, then $a \notin \mathfrak{m}$, and hence $\alpha = a(1 - \delta)$ with $\delta \in \mathfrak{m}\hat{A}$. It follows that

$$a^{-1}(1 + \delta + \delta^2 + ...)$$

is the inverse of α in $\hat{A}$. Hence this is a local ring. We already know that it is flat over A, and the faithful flatness follows from Corollary 2.20.

(c) Let $n \geq 1$. We have $\mathfrak{n}^n = \mathfrak{m}^n B$. Since the composition $A/\mathfrak{m}^n \to B/\mathfrak{n}^n B \to \hat{A}/\mathfrak{m}^n \hat{A}$ is an isomorphism, $B/\mathfrak{n}^n B \to \hat{A}/\mathfrak{m}^n \hat{A}$ is surjective. It remains to show that it is injective; that is, that $\mathfrak{m}^n \hat{A} \cap B = \mathfrak{n}^n B$. We have $B = A + \mathfrak{m}B = A + \mathfrak{m}^2 B = \cdots = A + \mathfrak{m}^n B$, so every element $b \in B$ can be written $b = a + \varepsilon$ with $a \in A$, $\varepsilon \in \mathfrak{n}^n B \subseteq \mathfrak{m}^n \hat{A}$. If, moreover, $b \in \mathfrak{m}^n \hat{A}$, then $a \in \mathfrak{m}^n \hat{A} \cap A = \mathfrak{m}^n$, so $b \in \mathfrak{n}^n B$.

Exercises 3.1. Is the usual topology on $\mathbb{R}$ defined by a subgroup filtration?

3.2. Show that $\mathbb{Z}_p$ is a principal local ring, with maximal ideal $p\mathbb{Z}_p$.

3.3. Let A be the ring of germs of real analytic functions in $0 \in \mathbb{R}$. Let $x \in A$ be the identity map on $\mathbb{R}$. Show that the map $f \mapsto \sum_n (f^{(n)}(0)/n!)T^n$ from A to $\mathbb{R}[[T]]$ is injective, and induces an isomorphism from the xA-adic completion of A onto $\mathbb{R}[[T]]$. Show that this is false if we replace A by the C^∞-functions in 0.

3.4. Find examples where $M \otimes_A \hat{A} \to \hat{M}$ is not surjective, or is not injective.

3.5. Let $B = \bigoplus_{n \geq 0} B_n$ be a graded ring. - (a) Show that B_0 is a subring of B, with unit. - (b) Show that if B is Noetherian, then so is B_0, and B is a finitely generated algebra over B_0.

3.6. Let A be an integral domain that is not a field, and let K be its field of fractions. Show that the conclusion of Proposition 3.11 does not hold if we take $M = K$, $N = A$, and $I = 0$.

3.7. Let $(A, \mathfrak{m})$ be a Noetherian local ring. Show that $\bigcap_{n\geq 0} \mathfrak{m}^n = 0$. Give a counter-example with A not Noetherian.

3.8. Let A be a Noetherian ring, and I, J ideals of A. Let $\hat{A}$ be the I-adic completion of A and $(A/J)\hat{}$ the completion of A/J for the $(I+J)/J$-adic topology. Show that there is a canonical isomorphism $\hat{A}/J\hat{A} \simeq (A/J)\hat{}$.

3.9. (nth root) Let $n \geq 2$ be an integer. Let $D = \mathbb{Z}[1/n]$. - (a) Consider the polynomial $S = (1+T)^n - 1 \in D[T]$. Show that $D[[S]] = D[[T]]$ and that there exists an $f(S) \in SD[[S]]$ such that $1 + S = (1 + f(S))^n$. - (b) Let A be a complete ring for the I-adic topology, where I is an ideal of A. Suppose that n is invertible in A. Let $x \in I$. Show that there exists a unique continuous homomorphism $\phi : D[[S]] \to A$ such that $\phi(S) = x$. Conclude that there exists a $y \in I$ such that $1 + x = (1 + y)^n$. - (c) Show, by giving an example, that statement (b) is false if n is not invertible in A.

3.10. Let k be a field of characteristic different from 2, and $A = k[x, y]/(y^2 - x^2(x + 1))$. Let $\hat{A}$ be the $\mathfrak{m}$-adic completion, where $\mathfrak{m} = (x, y)A$. Show that $\hat{A} \simeq k[[u, v]]/uv$.

3.11. Let A be a complete ring for the I-adic topology, where I is an ideal of A. Let $(M_n)_{n\geq 0}$ be A-modules such that $I^{n+1}M_n = 0$ and that there exist surjective homomorphisms $\pi_n : M_{n+1} \to M_n$ with $\mathrm{Ker}(\pi_n) = I^{n+1}M_{n+1}$. Let $M = \varprojlim M_n$ and denote the (surjective) canonical homomorphisms by $u_n : M \to M_n$. - (a) Fix $d \geq 0$. Show that for any $n \geq d$, there is an exact sequence

$$0 \to I^{d+1}M_n \to M_n \to M_d \to 0$$

- (b) Let us suppose that M_0 is generated over A by a finite number of elements $e_{0,1}, \dots, e_{0,m}$. Let $e_1, \dots, e_m \in M$ be such that $u_0(e_i) = e_{0,i}$, and define $\phi_n : A^m \to M_n$ by $(a_1, \dots, a_m) \mapsto \sum_i a_i u_n(e_i)$. Show that ϕ_n is surjective, and, using Lemma 3.1, that M is generated by the e_i. - (c) Let us moreover suppose that A is Noetherian and that M_0 is finitely generated over A. We are going to show that $\mathrm{Ker}\ u_n = I^{n+1}M$ (in other words, u_n induces an isomorphism $M/I^{n+1}M \simeq M_n$). (1) Let $K_n = \mathrm{Ker}\ u_n$. Show that $K_n = I^{n+1}M + K_{n+1}$. (2) Show that the canonical homomorphism $\varprojlim M/I^{n+1}M \to \varprojlim M_n$ is an isomorphism (use Corollary 3.14). (3) Apply Lemma 3.1 to the exact sequence of inverse systems

$$0 \to (K_n/I^{n+1}M)_n \to (M/I^{n+1}M)_n \to (M_n)_n \to 0$$

and show that $K_n = I^{n+1}M$.

3.12. Let A be a Noetherian ring, I an ideal of A, and $\hat{A}$ the I-adic completion of A. Let $a \in A$. Show that if a is not a zero divisor in A, then it is not a zero divisor in $\hat{A}$. However, show, by giving an example, that A can be an integral domain without $\hat{A}$ being one.

3.13. Let A be a Noetherian ring, I an ideal of A, and M a finitely generated A-module. Suppose that A is complete for the I-adic topology. Show that every submodule of M is closed.

3.14. Let A be a Noetherian ring, I an ideal of A, and $\hat{A}$ the formal I-adic completion of A. Show that the topology on $\hat{A}$ induced by that on A is the $I\hat{A}$-adic topology.

3.15. (Mittag–Leffler condition) Let $(A_n, \pi_n)_n$ be an inverse system of sets. For any pair $m > n$, let us denote the map $\pi_n \circ \pi_{n+1} \circ \cdots \circ \pi_{m-1} : A_m \to A_n$ by $\pi_{m,n}$. We say that the system $(A_n, \pi_n)_n$ satisfies the Mittag– Leffler condition if for every n, the descending sequence $(\pi_{m,n}(A_m))_{m>n}$ is stationary. - (a) Let $A'_n = \bigcap_{m>n} \pi_{m,n}(A_m)$. Show that $(A'_n, \pi_{n-1}|_{A'_n})_n$ is an inverse system and that the canonical map $\varprojlim A'_n \to \varprojlim A_n$ is bijective. - (b) Let us suppose that $(A_n, \pi_n)_n$ satisfies the Mittag–Leffler condition and that $A_n \neq \emptyset$ for all n. Show that $A'_n \neq \emptyset$, $A'_{n+1} \to A'_n$ is surjec- tive, and that $\varprojlim A'_n \neq \emptyset$. Deduce from this that $\varprojlim A_n \neq \emptyset$. - (c) Let $0 \to (A_n)_n \to (B_n)_n \xrightarrow{\rho} (C_n)_n \to 0$ be an exact sequence of inverse systems of Abelian groups such that $(A_n)_n$ satisfies the Mittag–Leffler condition. Let $(c_n)_n \in \varprojlim C_n$ and $X_n = \rho_n^{-1}(c_n)$, where $\rho_n : B_n \to C_n$ is the nth component of ρ. Show that $(X_n)_n$ is an inverse system of sets satisfying the Mittag–Leffler condition. Deduce from this that $\varprojlim B_n \to \varprojlim C_n$ is surjective. This gener- alizes Lemma 3.1.